DATA ANALYSIS REPORT

Project / Case Study: Bellabeat Smart Device Usage Analysis
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EXECUTIVE SUMMARY

This case study analyzed historical Fitbit smart-device data to identify consumer activity and sleep patterns and determine how these insights could inform Bellabeat's marketing strategy for its Time wellness watch. Using Python for data cleaning and analysis and Tableau for visualization, the study examined daily activity levels, activity intensity, hourly and weekly movement patterns, recorded sleep duration, and relationships between wellness measurements.

The analysis revealed considerable differences in participants' activity levels, with 57.14% of the 35 participants averaging fewer than 7,500 daily steps. Physical activity varied throughout the day and week, with average hourly steps peaking at 7:00 PM. Recorded sleep duration also varied across days of the week, while positive correlations between daily steps, activity intensity, and estimated calorie expenditure highlighted the value of monitoring multiple activity measurements.

Based on these findings, three marketing recommendations were developed: promote personalized activity goals, test routine-based activity messaging, and highlight comprehensive wellness tracking. These strategies could help Bellabeat communicate the benefits of its Time wellness watch, engage consumers with different activity habits, and develop more targeted marketing campaigns.

Although the findings provide useful insights into smart-device usage, the analysis relies on historical Fitbit data from 2016 and relatively small participant samples. Additional Bellabeat customer research and marketing experiments are recommended to validate these opportunities before broader implementation.

1. BUSINESS TASK

Business Problem

Bellabeat wants to better understand how consumers use smart devices and how patterns in smart-device usage could reveal growth opportunities. The company needs to determine how these consumer behaviors apply to Bellabeat customers and how to use them to strengthen the marketing strategy for one of its wellness products.

Business Objective

Analyze smart-device usage data to identify key consumer behavior trends, determine how those trends may apply to Bellabeat customers, and develop insights to guide Bellabeat's marketing strategy and support future growth.

Key Question(s)

- What are some trends in smart device usage?
- How could these trends apply to Bellabeat customers?
- How could these trends help influence Bellabeat marketing strategy?

Stakeholders

The primary stakeholders are Bellabeat cofounder and Chief Creative Officer **Urška Sršen**, cofounder and executive team member **Sando Mur**, the Bellabeat executive team, and the Bellabeat marketing analytics team. These stakeholders will use the analysis and recommendations to better understand consumer smart-device behavior and make informed decisions about Bellabeat's marketing strategy.

2. DATA SOURCES

Time Period 1: March 12 – April 11, 2016

Data Source 1 — dailyActivity_merged

Source: FitBit Fitness Tracker Data (Kaggle, Möbius) — Fitabase Data 3.12.16–4.11.16
Format: CSV, imported into Google Sheets
Time Period: March 12 – April 12, 2016 (observed dates).
Size: 457 rows, 15 columns, and 35 unique participant IDs.
Description: Contains daily physical activity measurements, including total steps, distance traveled, calories burned, and time spent at different activity intensity levels. This dataset is useful for understanding daily exercise habits, physical activity levels, and sedentary behavior among Fitbit users.

Data Source 2 — hourlySteps_merged

Source: FitBit Fitness Tracker Data (Kaggle, Möbius) — Fitabase Data 3.12.16–4.11.16
Format: CSV, imported into Google Sheets
Time Period: March 12 – April 12, 2016
Size: 24,084 rows, 3 columns, and 34 unique participant IDs.
Description: Records the number of steps taken by each participant during individual hours. This dataset is useful for identifying patterns in physical activity throughout the day, including periods when users tend to be more or less active.

Data Source 3 — hourlyIntensities_merged

Source: FitBit Fitness Tracker Data (Kaggle, Möbius) — Fitabase Data 3.12.16–4.11.16
Format: CSV, imported into Google Sheets

Time Period: March 12 – April 12, 2016
Size: 24,084 rows, 4 columns, and 34 unique participant IDs.
Description: Contains hourly activity intensity measurements, including total and average intensity. This dataset helps examine changes in physical activity intensity throughout the day and provides additional context for understanding users' exercise and movement patterns.

Data Source 4 — hourlyCalories_merged

Source: FitBit Fitness Tracker Data (Kaggle, Möbius) — Fitabase Data 3.12.16–4.11.16
Format: CSV, imported into Google Sheets
Time Period: March 12 – April 12, 2016
Size: 24,084 rows, 3 columns, and 34 unique participant IDs.
Description: Records estimated calories burned by participants during each hour. This dataset is useful for examining patterns in energy expenditure throughout the day and exploring how calorie expenditure varies alongside physical activity.

Data Source 5 — minuteSleep_merged

Source: FitBit Fitness Tracker Data (Kaggle, Möbius) — Fitabase Data 3.12.16–4.11.16
Format: CSV, imported into Google Sheets
Time Period: March 12 – April 12, 2016
Size: 198,559 rows, 4 columns, and 23 unique participant IDs.
Description: Contains minute-level sleep observations, including timestamps, recorded sleep-state values, and sleep log identifiers. This dataset can be used to examine sleeping patterns, sleep timing, and potential sleep duration after validating the recorded sleep states and logs.

Data Source 6 — heartrate_seconds_merged

Source: FitBit Fitness Tracker Data (Kaggle, Möbius) — Fitabase Data 3.12.16–4.11.16
Format: CSV, imported into Google Sheets
Time Period: March 12 – April 12, 2016
Size: 1,154,681 rows, 3 columns, and 14 unique participant IDs.
Description: Contains heart-rate measurements recorded at intervals of several seconds, along with participant identifiers and timestamps. This dataset is useful for exploring heart-rate variations and supplementing physical activity analysis with physiological measurements.

Time Period 2: April 12 – May 12, 2016

Data Source 7 — dailyActivity_merged

Source: FitBit Fitness Tracker Data (Kaggle, Möbius) — Fitabase Data 4.12.16–5.12.16
Format: CSV, imported into Google Sheets.
Time Period: April 12 – May 12, 2016.
Size: 940 rows, 15 columns, and 33 unique participant IDs.
Description: Contains daily measurements of steps, distance, calories burned, and time spent in different activity levels. This dataset provides additional observations of daily physical activity

and can be combined with the corresponding dataset from the first period after checking for overlapping records.

Data Source 8 — hourlySteps_merged

Source: FitBit Fitness Tracker Data (Kaggle, Möbius) — Fitabase Data 4.12.16–5.12.16
Format: CSV, imported into Google Sheets.
Time Period: April 12 – May 12, 2016.
Size: 22,099 rows, 3 columns, and 33 unique participant IDs.
Description: Records hourly step counts for each participant. This dataset is useful for examining hourly activity patterns during the second period and extending the analysis of when users are most physically active.

Data Source 9 — hourlyIntensities_merged

Source: FitBit Fitness Tracker Data (Kaggle, Möbius) — Fitabase Data 4.12.16–5.12.16
Format: CSV, imported into Google Sheets.
Time Period: April 12 – May 12, 2016.
Size: 22,099 rows, 4 columns, and 33 unique participant IDs.
Description: Contains total and average activity intensity measurements for each hour. This dataset helps identify differences in physical activity intensity throughout the day and supports comparisons with the corresponding intensity measurements from the first period.

Data Source 10 — hourlyCalories_merged

Source: FitBit Fitness Tracker Data (Kaggle, Möbius) — Fitabase Data 4.12.16–5.12.16
Format: CSV, imported into Google Sheets.
Time Period: April 12 – May 12, 2016.
Size: 22,099 rows, 3 columns, and 33 unique participant IDs.
Description: Records estimated hourly calorie expenditure for each participant. This dataset provides additional information about energy expenditure and allows the analysis of calorie-burning patterns across the two collection periods.

Data Source 11 — minuteSleep_merged

Source: FitBit Fitness Tracker Data (Kaggle, Möbius) — Fitabase Data 4.12.16–5.12.16
Format: CSV, imported into Google Sheets.
Time Period: April 12 – May 12, 2016.
Size: 188,521 rows, 4 columns, and 24 unique participant IDs.
Description: Contains minute-level sleep observations, recorded sleep-state values, timestamps, and sleep log identifiers. This dataset provides additional sleep records that can help identify patterns in sleeping behavior, sleep timing, and duration after the records are validated and prepared for analysis.

Data Source 12 — heartrate_seconds_merged

Source: FitBit Fitness Tracker Data (Kaggle, Möbius) — Fitabase Data 4.12.16–5.12.16
Format: CSV, imported into Google Sheets.
Time Period: April 12 – May 12, 2016.

Size: 2,483,658 rows, 3 columns, and 14 unique participant IDs.
Description: Contains heart-rate measurements recorded at second-level intervals, along with participant identifiers and timestamps. This dataset can be used to examine heart-rate patterns during the second collection period and provide additional physiological context for the physical activity analysis.

Data Reliability and Limitations

Data Credibility and Ownership

The datasets were obtained from Kaggle's FitBit Fitness Tracker Data collection, published by Möbius. The original data were collected as part of a research project by Furberg et al. (2016), involving Fitbit users recruited through Amazon Mechanical Turk. The Kaggle dataset is listed under the CC0: Public Domain license, allowing its use and modification for this educational analysis.

Sample Size and Representation

The original dataset documentation describes 30 consenting Fitbit users. However, the initial data profiling identified differences in the number of unique participant IDs across datasets, including 35 IDs in the first period's daily activity dataset and 33 in the second. This discrepancy requires further investigation.
Additionally, participants were recruited through a nonrandom sampling method, so the findings may not accurately represent the general population of smart-device users. Because the data come from Fitbit users rather than Bellabeat customers, any application of the findings to Bellabeat's target audience should be considered exploratory.

Data Completeness and Quality

Initial profiling identified no missing cells in the 12 selected datasets. However, the number of observations and unique participant IDs varies across datasets, indicating that not every participant has the same amount of recorded activity, sleep, or heart-rate information.
Duplicate records were also identified in the minuteSleep datasets, with 525 flagged duplicates in the first period and 543 in the second. Additionally, the two collection periods contain overlapping dates, creating a potential risk of double-counting observations when combining the datasets.
These issues will require further investigation and appropriate handling during the Process Phase.

Data Collection and Measurement Limitations

The datasets contain information collected during approximately two months in 2016, limiting their usefulness for understanding current or long-term smart-device usage trends.
The measurements also rely on Fitbit devices, which may contain inaccuracies caused by tracking limitations, device usage, or periods when participants were not wearing their devices.
Furthermore, the selected datasets do not include demographic information such as age or gender, making it difficult to examine how usage patterns differ across consumer groups.

Privacy and Ethical Considerations

Although the datasets use numerical participant identifiers rather than personal names, detailed activity, sleep, and heart-rate measurements may still contain potentially sensitive information. For this reason, results should primarily be presented through aggregated statistics and visualizations rather than individual participant records.

Overall Assessment

Despite these limitations, the datasets provide useful information for exploring historical smart-device usage patterns, including physical activity, sedentary behavior, calorie expenditure, sleep, and heart rate. The analysis can help identify consumer behavior trends and generate potential marketing insights for Bellabeat. However, the findings should not be generalized to all smart-device users or Bellabeat customers without additional supporting evidence.

References

Möbius. FitBit Fitness Tracker Data. Kaggle. https://www.kaggle.com/datasets/arashnic/fitbit?resource=download-directory
Furberg, R., Brinton, J., Keating, M., & Ortiz, A. (2016). Crowd-sourced Fitbit datasets 03.12.2016–05.12.2016. Zenodo. https://doi.org/10.5281/zenodo.53894

3. DATA CLEANING AND MANIPULATION

Tools Used

Python (Pandas, NumPy) and Jupyter Notebook were used to inspect, clean, consolidate, validate, and export the datasets. Python was used to identify missing values, duplicate observations, inconsistent measurements, and potential outliers. Jupyter Notebook was used to organize the cleaning procedures, document decisions, and verify results.

Cleaning Process

- **Removed duplicate records:** Inspected all six data types for duplicate observations within and across the two collection periods. Identical overlapping records were resolved by prioritizing Period 2 while preserving nonoverlapping observations. For the minute sleep datasets, 1,068 exact duplicates were removed within the original datasets, followed by 3,232 overlapping participant-minute observations. The heart rate datasets contained 23,424 overlapping records, while each hourly dataset contained 175 overlapping records.
- **Handled missing or null values:** Examined each dataset for missing values, incomplete observations, and recording gaps. The daily activity dataset contained 133 observations with zero steps, including six with zero calories. The six zero-calorie observations were excluded from the primary daily calorie analysis but retained in the consolidated dataset. Missing hourly observations and irregular heart rate recording intervals were documented rather than replaced with estimated values.
- **Corrected data types and formatting:** Converted date and timestamp columns into appropriate datetime formats and verified that numerical measurements were stored

using suitable data types. Participant identifiers, activity measurements, sleep-state values, and heart rate observations were checked for consistency. Date and hour variables were also extracted to support future analysis.

- **Standardized column names and categories:** Confirmed that corresponding datasets from both collection periods had matching column names and structures before consolidation. Standardized source-period labels and created consistent date, time, and quality-flag variables to support comparisons across datasets.
- **Investigated invalid values and outliers:** Examined unusual activity measurements, extreme heart rates, and irregular sleep durations. The hourly datasets contained 134 observations exceeding 5,000 steps per hour, 752 observations exceeding 300 estimated calories per hour, and 20,027 observations with zero recorded intensity. The heart rate dataset contained readings ranging from 36 to 203 BPM, including 168 unusually rapid changes. Sleep inspections identified 123 logs shorter than 120 minutes and nine longer than 840 minutes. Potentially unusual measurements were generally retained and flagged rather than automatically removed.
- **Created calculated fields and new variables:** Created variables for activity dates, hours of the day, recording intervals, and changes in heart rate. Quality flags were developed to identify unusual measurements and incomplete recording periods. For the minute sleep data, individual observations were summarized into 1,007 sleep logs, including recorded duration, sleep-state totals, sleep-state percentages, and indicators of unusual sleep duration.
- **Joined and merged datasets:** Consolidated corresponding datasets from the two collection periods while resolving overlapping observations. The hourly steps, intensities, and calorie datasets were compared using participant identifiers and timestamps, confirming that all 46,008 participant-hour observations aligned. Sleep observations were grouped using participant and sleep-log identifiers. The cleaned datasets were exported separately to preserve their measurement levels and support future analysis.

Quality Checks

Data quality was verified throughout the cleaning process using Python validation checks. These checks examined missing essential values, duplicate observations, invalid timestamps, numerical measurements, and consistency between calculated variables and original data.

Following consolidation, the daily activity dataset contained 1,373 unique participant-date observations, with 1,367 retained for the primary calorie analysis. Each hourly dataset contained 46,008 unique participant-hour observations. The minute sleep dataset contained 382,780 participant-minute observations summarized into 1,007 sleep logs, while the heart rate dataset contained 3,614,915 unique participant-timestamp observations across 15 participants.

Additional checks confirmed that no duplicate observations remained within the cleaned datasets. The hourly datasets were aligned, sleep-state totals matched recorded durations, and heart rate quality flags were validated. Known recording gaps and unusual measurements were documented to preserve transparency and avoid introducing unsupported assumptions.

Finally, the cleaned datasets were exported as CSV files and reloaded to verify their dimensions, essential values, and calculated fields. These checks confirmed that the datasets were successfully processed and prepared for the Analyze phase, while retaining documented limitations that must be considered when interpreting the results.

4. ANALYSIS

Approach

Python, Pandas, and NumPy were used to analyze the cleaned and consolidated Fitbit datasets. Descriptive statistics, grouped comparisons, participant-level summaries, and correlation analyses were performed to identify trends and relationships in physical activity, estimated calorie expenditure, and recorded sleep duration.

Mean, median, standard deviation, minimum, and maximum values were calculated to summarize the distributions of key measurements. Activity and sleep patterns were examined by hour of the day, day of the week, and participant activity level.

Pearson and Spearman correlations were used to investigate relationships between numerical measurements. Participant-level comparisons and sensitivity analyses were also performed to evaluate how repeated observations, unusual values, and recording limitations influenced the results.

These methods were appropriate for identifying smart-device usage patterns and exploring potential applications to Bellabeat's wellness products and marketing strategy.

Analysis Steps

1. **Import and Inspect Cleaned Datasets:** Imported the cleaned Fitbit datasets into Python, verified their dimensions and column names, inspected data types, and converted date and timestamp variables into appropriate formats.
2. **Calculate Descriptive Statistics:** Calculated the mean, median, standard deviation, minimum, and maximum for daily steps, activity intensity, sedentary minutes, estimated calorie expenditure, recorded sleep duration, and heart rate.
3. **Evaluate Unusual Measurements:** Examined quality flags for unusual sleep durations and heart-rate measurements. Conducted sensitivity analyses to compare descriptive statistics using complete and unflagged observations.
4. **Identify Daily and Hourly Usage Trends:** Analyzed physical activity by day of the week and hour of the day. Examined hourly steps, activity intensity, estimated calorie expenditure, and recorded sleep duration to identify patterns throughout the day and week.
5. **Compare Weekday and Weekend Behavior:** Compared daily steps and recorded sleep duration between weekdays and weekends. Performed participant-level comparisons to examine individual differences while reducing the influence of unequal observation counts.
6. **Classify and Compare Participant Activity Levels:** Grouped 35 participants into four categories based on average daily step counts. Compared active minutes, sedentary time, and estimated calorie expenditure across the activity groups.

7. **Examine Relationships Between Variables:** Calculated Pearson and Spearman correlations between daily steps, activity intensity, sedentary minutes, and estimated calories. Matched daily activity with sleep records to investigate additional relationships. Conducted participant-level correlation analyses and time-accounting checks to evaluate potential measurement-related effects.
8. **Summarize and Interpret Key Findings:** Identified the strongest activity, sleep, and calorie-expenditure findings. Connected these results to Bellabeat's business questions and explored potential applications involving personalized activity goals, wellness reminders, sleep tracking, and individual progress monitoring.
9. **Export and Validate Analytical Results:** Exported 10 analytical summary tables as CSV files for visualization and reporting. Reloaded and verified the exported files to confirm that their row counts, column names, and values matched the original analytical outputs.

Results

Daily Activity and Participant Differences

Across 1,373 participant-day observations, average daily steps were 7,377.38, indicating considerable variation in recorded physical activity.
Of the 35 participants, 57.14% averaged fewer than 7,500 daily steps, while 20% averaged at least 10,000 steps. Participants with higher average step counts generally recorded more very active minutes, although sedentary time and estimated calorie expenditure did not consistently follow the same pattern.

Hourly and Weekly Activity Trends

Physical activity was lowest overnight and generally increased throughout the morning. Average hourly steps were highest at 7 PM (554.89 steps), while activity intensity and estimated calorie expenditure also reached relatively high levels during early evening hours.
Participant-level comparisons showed approximately 206 fewer steps on weekends than weekdays, on average. However, individual activity patterns varied considerably.

Recorded Sleep Patterns

The initial analysis of 1,007 sleep logs found an average recorded duration of 6.34 hours. After excluding 132 logs flagged for unusual durations, the average increased to 6.93 hours across 875 unflagged logs.
Sunday had the longest average recorded sleep-log duration at 7.36 hours, while Saturday had the shortest at 6.68 hours. Participant-level comparisons suggested slightly longer recorded sleep durations on weekends, although differences were small.

Relationships Between Activity and Calorie Expenditure

Daily steps showed a moderate positive correlation with estimated calorie expenditure (Pearson = 0.576).

The relationship was stronger when examining day-to-day changes within participants (Pearson = 0.802), suggesting that higher-than-usual step counts generally corresponded with higher-than-usual estimated calorie expenditure for the same participant.

Activity and Sleep Relationships

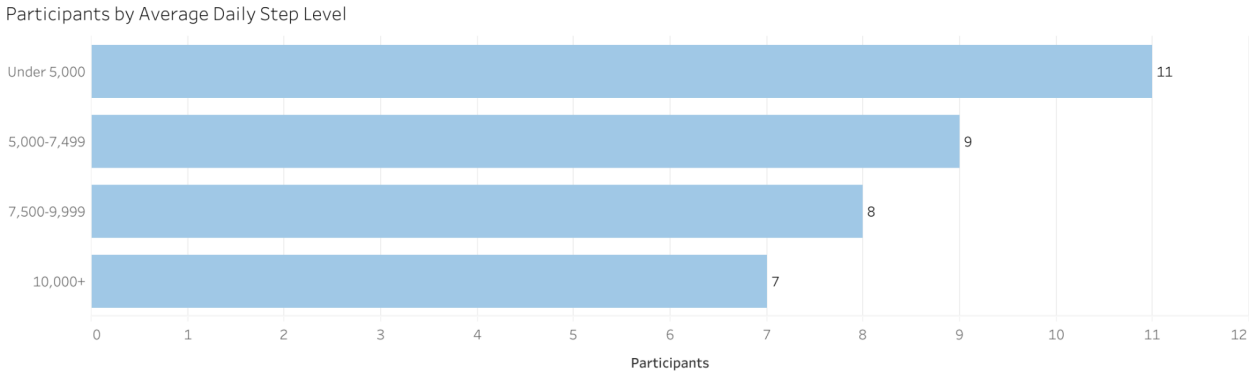
Most activity measurements showed weak correlations with recorded sleep-log duration. Sedentary minutes initially showed a stronger negative relationship with sleep duration, but additional analysis suggested that daily time accounting and recording coverage may contribute to this association.
Therefore, the sedentary-and-sleep relationship was not treated as a primary behavioral finding.

Overall Interpretation

The findings highlight differences in activity levels between participants, relatively high recorded movement during early evening hours, and the potential importance of tracking individual activity progress over time.
These results suggest possible opportunities for Bellabeat to explore personalized activity goals, routine-based reminders, sleep tracking, and individual progress monitoring.
However, the analysis uses historical Fitbit data from a limited sample rather than Bellabeat customer data. The findings are exploratory and should be validated with additional customer research before informing marketing decisions.

5. SUPPORTING VISUALIZATIONS AND KEY FINDINGS

Figure 1. Participants by Average Daily Step Level

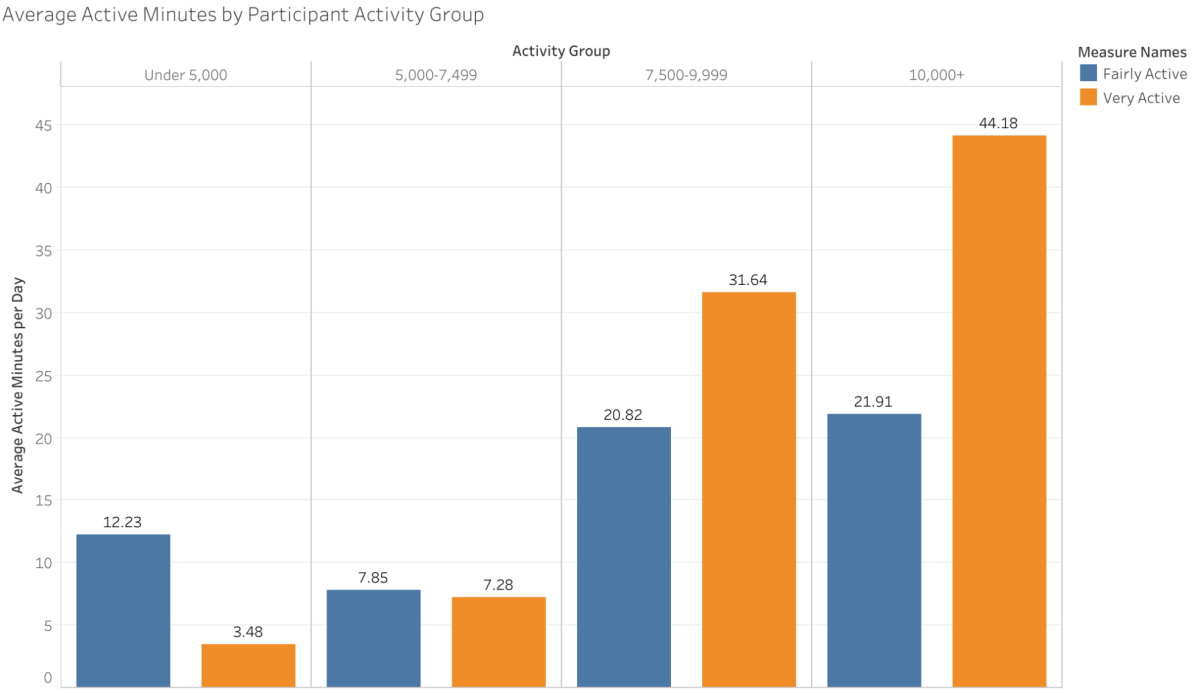


Key Finding

- The 35 participants represented in the activity-group analysis were distributed across four daily step categories: Under 5,000 (11 participants), 5,000–7,499 (9 participants), 7,500–9,999 (8 participants), and 10,000+ (7 participants).
- The largest group consisted of participants averaging fewer than 5,000 daily steps, accounting for 31.43% of the sample.
- Only 20% of participants averaged at least 10,000 daily steps, indicating that higher daily step counts were less common within this sample.

- These differences suggest an opportunity for Bellabeat to explore personalized activity goals and engagement strategies based on users' existing activity levels.

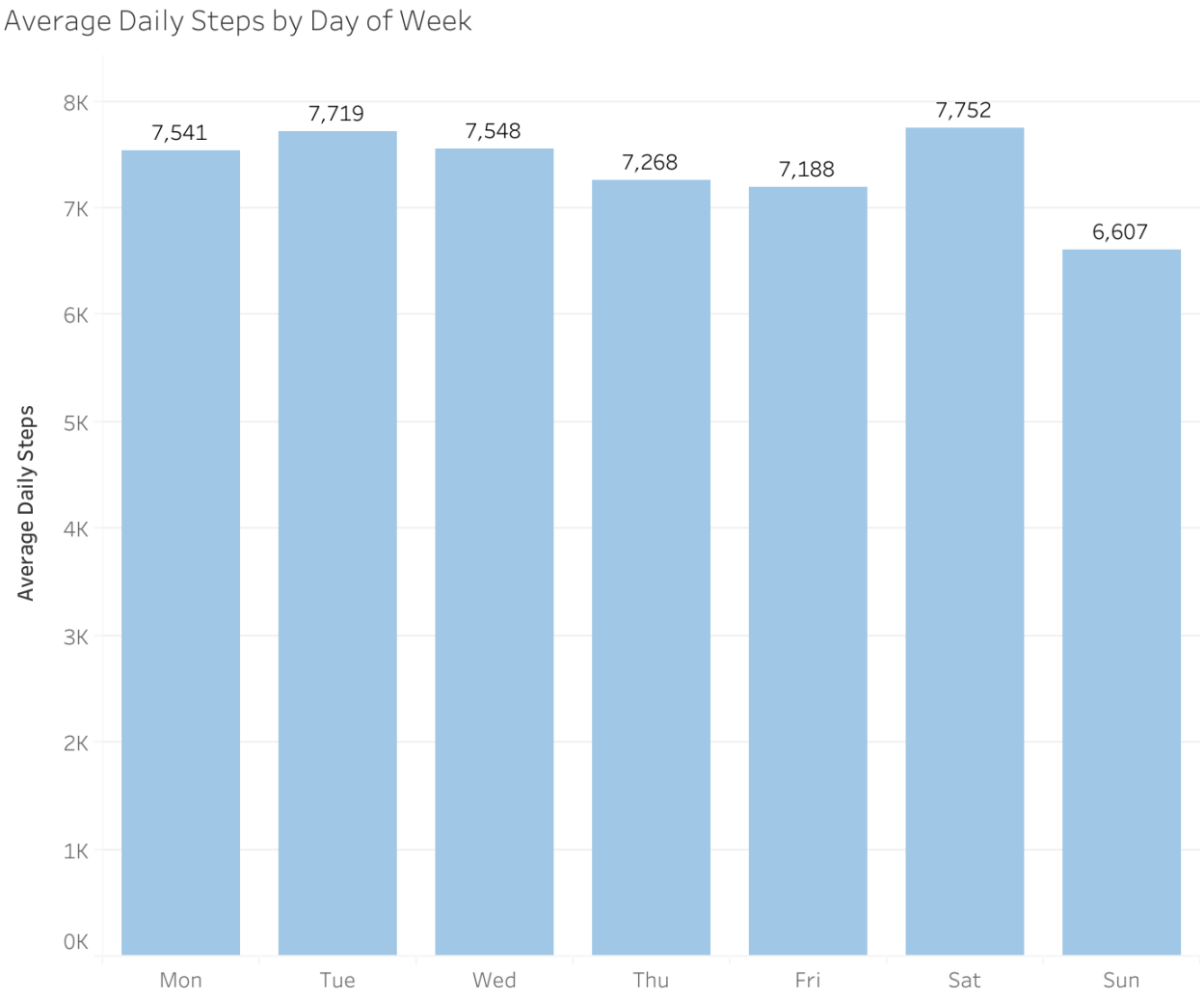
Figure 2. Average Active Minutes by Participant Activity Group



Key Finding

- Participants in the highest daily step category generally recorded substantially more active minutes than those in lower step categories.
- Average very active minutes increased from 3.48 minutes per day in the Under 5,000 group to 44.18 minutes per day in the 10,000+ group.
- The 10,000+ group also recorded the highest average fairly active minutes, at 21.91 minutes per day, compared with 12.23 minutes in the Under 5,000 group.
- These findings indicate that higher daily step counts were associated with greater amounts of higher-intensity physical activity, supporting the potential value of tracking both step counts and activity intensity in Bellabeat Time.

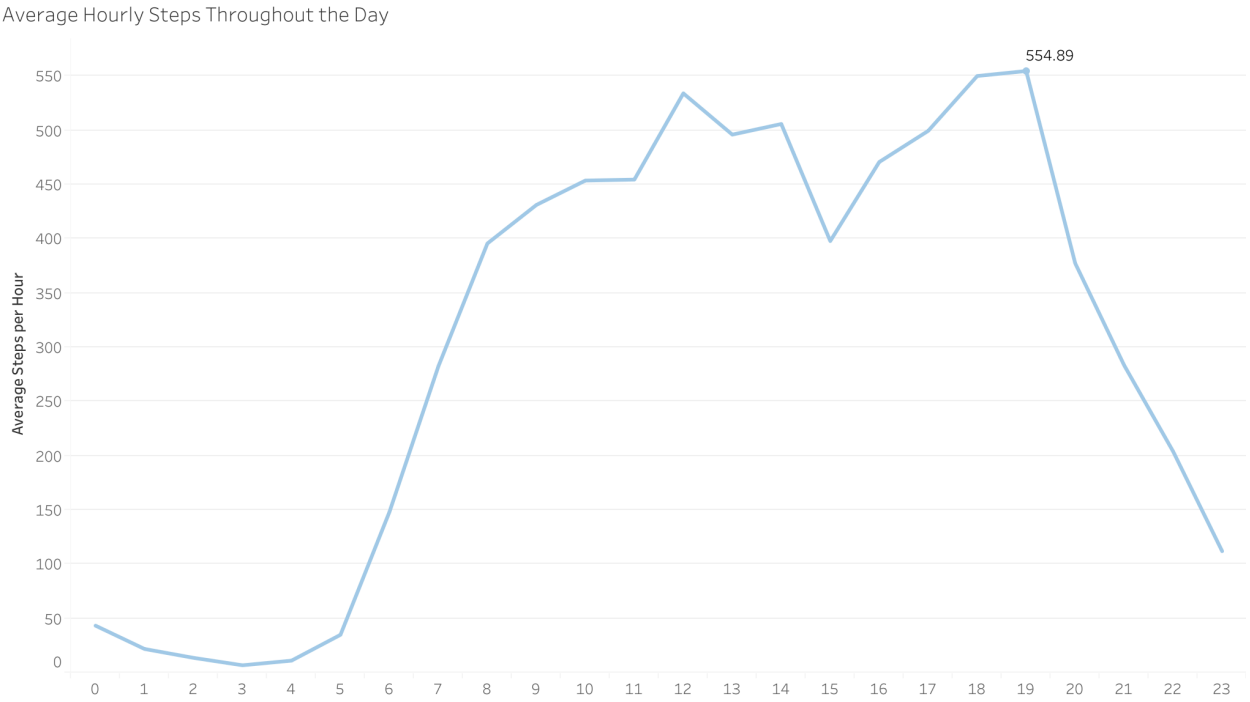
Figure 3. Average Daily Steps by Day of Week



Key Finding

- Average daily steps varied throughout the week, with Saturday recording the highest average at 7,752 steps and Sunday recording the lowest at 6,607 steps.
- Tuesday recorded the second-highest average at 7,719 steps, while Friday had one of the lower averages at 7,188 steps.
- The difference between Saturday and Sunday was approximately 1,146 steps, suggesting that participants' physical activity patterns differed between the two weekend days.
- These patterns suggest that Bellabeat could investigate day-specific activity reminders or weekly step goals to encourage consistent movement throughout the week.

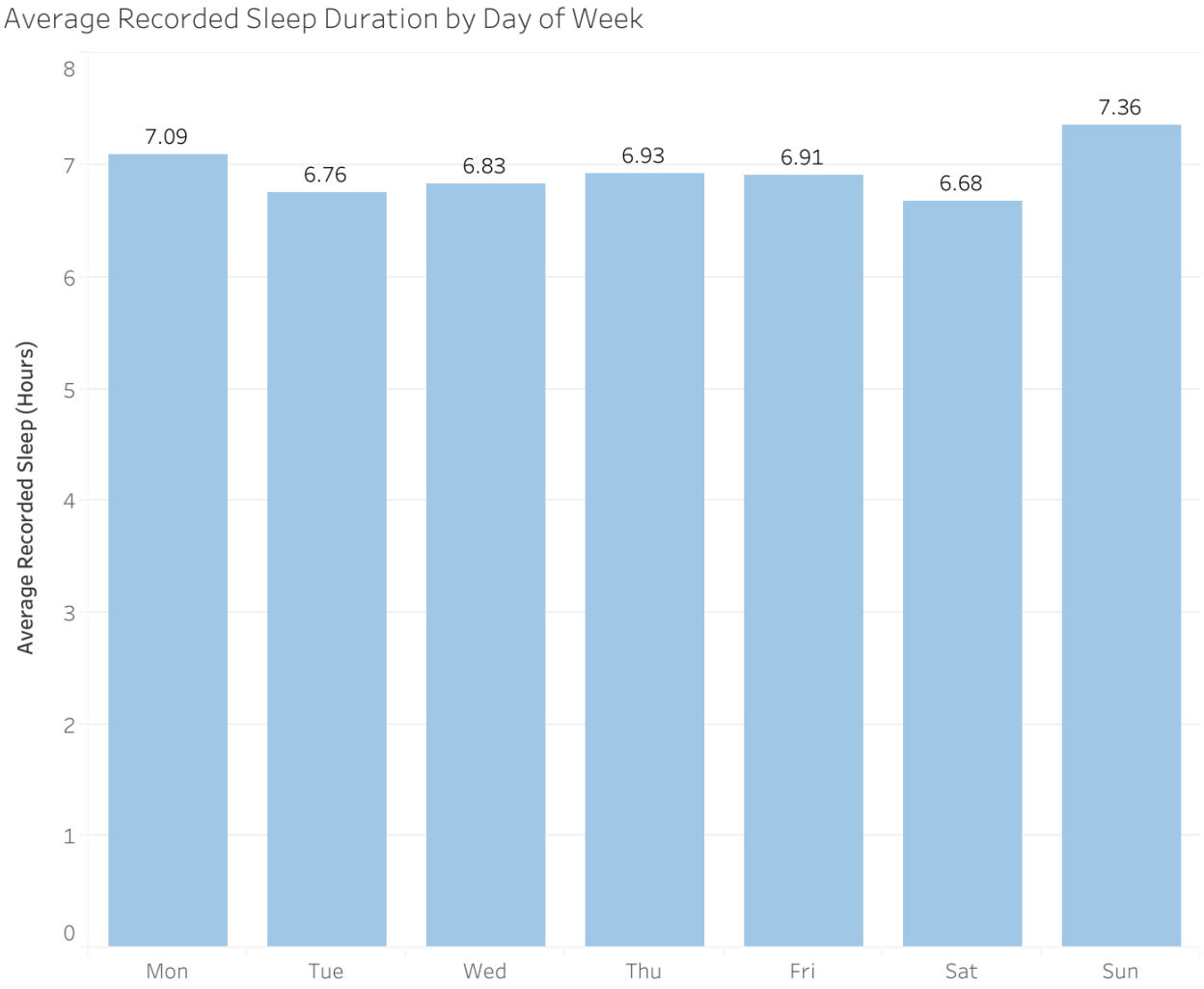
Figure 4. Average Hourly Steps Throughout the Day



Key Finding

- Hourly step activity was generally lowest during overnight and early-morning hours, followed by a substantial increase beginning in the morning.
- Activity remained relatively elevated throughout much of the daytime and early evening, reaching its highest recorded hourly average of approximately 554.89 steps at 7:00 PM (19:00).
- Step activity declined noticeably after the evening peak, indicating a clear daily pattern of movement within the recorded observations.
- These findings suggest that Bellabeat could explore time-based activity reminders or personalized movement goals that account for users' typical daily routines.

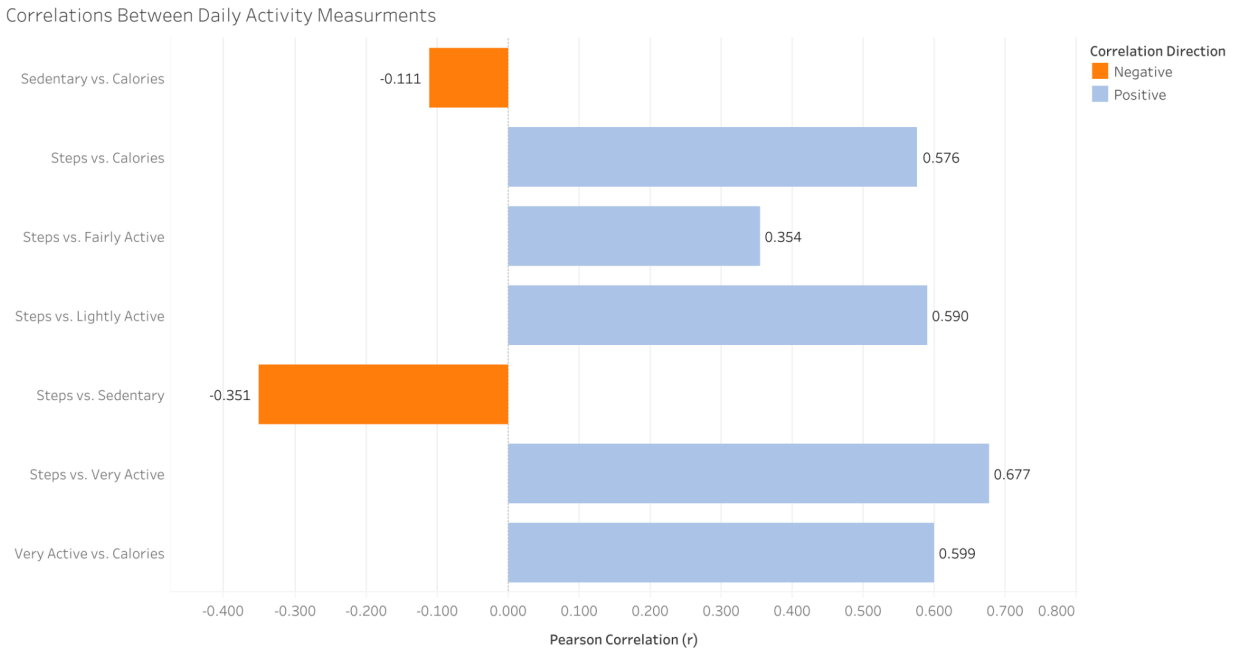
Figure 5. Average Recorded Sleep Duration by Day of Week



Key Finding

- Among the 875 unflagged sleep logs, average recorded sleep-log duration ranged from approximately 6.68 to 7.36 hours across the seven days of the week.
- Sunday recorded the longest average sleep duration at 7.36 hours, while Saturday recorded the shortest at 6.68 hours.
- The difference between the highest and lowest daily averages was approximately 0.68 hours, or 41 minutes, indicating some variation in recorded sleep duration throughout the week.
- These findings suggest an opportunity for Bellabeat to investigate sleep consistency and explore features such as sleep-tracking summaries, bedtime reminders, and personalized sleep insights.

Figure 6. Correlations Between Daily Activity Measurements



Key Finding

- Daily steps demonstrated a positive correlation with very active minutes (Pearson r = 0.677), indicating that days with higher recorded step counts generally corresponded to days with more vigorous physical activity.
- Daily steps were also positively correlated with calories burned (r = 0.576), while very active minutes showed a positive correlation with calories burned (r = 0.599).
- Daily steps had a negative correlation with sedentary minutes (r = -0.351), suggesting that higher step counts tended to be associated with less sedentary time.
- The correlation between sedentary minutes and calories burned was weak and negative (r = -0.111), indicating a limited linear relationship between those measurements.
- These relationships highlight the potential value of combining step counts, activity intensity, sedentary time, and calorie estimates when evaluating physical activity patterns. However, correlations represent statistical associations and do not establish causation.

Overall Key Findings

- **Variation in Daily Activity Levels:** Participants demonstrated different daily step patterns, with 31.43% of the activity-group sample averaging fewer than 5,000 steps and 20% averaging at least 10,000 steps. This highlights the potential importance of personalized activity goals.
- **Relationship Between Steps and Activity Intensity:** Participants in higher step categories generally recorded more very active minutes, suggesting that step counts and activity intensity can provide complementary information about physical activity behavior.

- **Weekly Activity Patterns:** Average daily steps varied throughout the week, with Saturday recording the highest average and Sunday the lowest. These differences suggest that activity engagement may benefit from strategies tailored to daily routines.
- **Daily Activity Timing:** Hourly steps increased during the morning and remained relatively elevated throughout the day, reaching a peak around 7:00 PM. These patterns could help inform the timing of personalized activity reminders.
- **Sleep Duration Patterns:** Average recorded sleep duration ranged from 6.68 to 7.36 hours across the week, with Sunday recording the longest average duration and Saturday the shortest. This supports further exploration of sleep consistency and sleep-related wellness features.
- **Relationships Between Activity Measurements:** Higher step counts were positively associated with very active minutes and calories burned, while showing a negative association with sedentary minutes. These findings support examining multiple activity metrics rather than relying exclusively on daily step totals.
- **Implications for Bellabeat Time:** The combined findings suggest opportunities to explore personalized activity goals, movement reminders, activity-intensity tracking, and sleep insights. Such features could help Bellabeat encourage consistent engagement with physical activity and wellness monitoring.

Data Interpretation Note: These findings are based on historical Fitbit activity and sleep records collected in 2016. The datasets contain relatively small samples, and participant coverage varies across measurements. Results should therefore be treated as exploratory and should not be assumed to represent all fitness-tracker users or Bellabeat customers.

6. TOP THREE RECOMMENDATIONS

Recommendation 1 — Promote Personalized Activity Goals

Recommendation:
Develop targeted marketing content highlighting how Bellabeat Time can support individual activity monitoring and progress tracking. Emphasize realistic, personalized movement goals that accommodate users with different fitness levels, rather than promoting a single universal daily step target.

Evidence:
Among the 35 participants analyzed, 57.14% averaged fewer than 7,500 daily steps, while only 20% averaged at least 10,000 steps. Participants with higher daily step counts also generally recorded more higher-intensity physical activity. Average very active minutes increased from 3.48 minutes in the Under 5,000 step group to 44.18 minutes in the 10,000+ step group.

Expected Benefit:
Personalized activity messaging could help Bellabeat appeal to consumers with different fitness levels, make its marketing more relevant, and encourage interest in consistent activity tracking and individual progress monitoring.

Recommendation 2 — Test Routine-Based Activity Messaging

Recommendation:
Develop and test marketing campaigns emphasizing movement throughout the day and week. Explore time-specific wellness content, activity reminders, and flexible weekly movement goals that encourage consumers to incorporate physical activity into their existing routines.

Evidence:
Average hourly steps were highest at 7:00 PM, reaching approximately 554.89 steps per hour, while overnight activity was considerably lower. Average daily steps also varied across the week, with Saturday recording the highest average of approximately 7,752 steps and Sunday the lowest at 6,607 steps. These findings indicate variation in recorded activity by time of day and day of week.

Expected Benefit:
Routine-based messaging could help Bellabeat identify effective opportunities to engage potential customers, encourage consistent wellness habits, and improve the relevance of its digital marketing campaigns. Testing different messages and delivery times would help determine which approaches are most effective.

Recommendation 3 — Highlight Comprehensive Wellness Tracking

Recommendation:
Develop marketing content emphasizing Bellabeat Time's ability to support broader wellness awareness through activity and sleep monitoring. Highlight the value of examining multiple wellness measurements, including daily steps, activity intensity, and sleep patterns, rather than focusing exclusively on daily step counts.

Evidence:
Average recorded sleep duration varied from approximately 6.68 hours on Saturday to 7.36 hours on Sunday. Daily steps demonstrated positive correlations with very active minutes ($r = 0.677$) and estimated calories burned ($r = 0.576$). These results suggest that different measurements provide complementary information about activity and wellness patterns, although the correlations do not establish causal relationships.

Expected Benefit:
Emphasizing multiple wellness measurements could strengthen Bellabeat Time's positioning as a comprehensive wellness-tracking product. This approach may help attract consumers interested in understanding their daily habits, monitoring personal progress, and developing greater awareness of their overall wellness.

CONCLUSION

This analysis examined historical Fitbit smart-device data to identify consumer activity and sleep patterns and explore how these findings could inform Bellabeat Time's marketing strategy.

The results revealed considerable differences in daily activity levels among participants, identifiable hourly and weekly movement patterns, and variation in recorded sleep duration. Additionally, positive relationships between daily steps, activity intensity, and estimated calorie expenditure demonstrated the potential value of examining multiple activity measurements.

Based on these findings, three recommendations were developed: promote personalized activity goals, test routine-based activity messaging, and highlight comprehensive wellness tracking. Together, these strategies could help Bellabeat communicate the benefits of its Time wellness watch, engage consumers with different activity habits, and develop more targeted marketing content.

However, the analysis relies on historical Fitbit data from 2016, with relatively small participant samples and varying data coverage. The findings are exploratory and should not be assumed to represent Bellabeat customers. Additional customer research and marketing experiments are recommended before implementing these strategies on a larger scale.

Overall, the analysis provides a data-informed foundation for exploring marketing opportunities that emphasize personalized progress, consistent activity habits, and broader wellness awareness.

APPENDIX

Appendix A — Supporting Analysis Tables

This section presents supplementary summary tables generated during the analysis of Fitbit activity and sleep data. These tables provide additional numerical evidence supporting the findings and visualizations discussed in the main report.

Table A.1 — Participant Distribution by Average Daily Step Level

ActivityGroup	Participants	AverageSteps	MedianSteps	MedianRecordedDays	MinRecordedDays	MaxRecordedDays	Percentage
Under 5,000	11	2,897.62	2,944.50	40	8	62	31.43%
5,000-7,499	9	6,418.93	6,616.93	42	30	49	25.71%
7,500-9,999	8	8,657.36	8,623.35	41.5	32	45	22.86%
10,000+	7	12,694.27	11,732.97	42	37	49	20%

Table A.1 Description: This table summarizes the distribution of 35 Fitbit participants across four average daily step categories. The largest group averaged fewer than 5,000 steps per day (31.43%), while 20% averaged at least 10,000 steps. These results provide additional numerical support for Figure 1 and highlight differences in physical activity levels among participants.

Table A.2 — Activity and Calorie Metrics by Participant Activity Group

ActivityGroup	Participants	AvgVeryActiveMinutes	AvgFairlyActiveMinutes	AvgLightlyActiveMinutes	AvgSedentaryMinutes	AvgDailyCalories
Under 5,000	11	3.48	12.23	117.15	1,171.23	2,122.47
5,000-7,499	9	7.28	7.85	226.68	1,043.57	1,959.42
7,500-9,999	8	31.64	20.82	191.34	813.16	2,739.38
10,000+	7	44.18	21.91	230.4	950.38	2,436.13

Table A.2 Description: This table compares average daily activity intensity across four participant activity groups. Participants averaging at least 10,000 daily steps recorded the highest average very active minutes (44.18 minutes), compared with 3.48 minutes among participants averaging fewer than 5,000 steps. The highest step group also recorded more fairly active and lightly active minutes than the lowest step group. These findings provide additional numerical support for Figure 2 and highlight differences in physical activity intensity across participant activity levels.

Table A.3 — Average Daily Steps by Day of Week

DayOfWeek	Observations	AverageSteps	MedianSteps	StdSteps
Monday	188	7,541.32	7,420.00	4,995.20
Tuesday	201	7,718.96	8,135.00	4,799.81
Wednesday	198	7,547.58	7,351.00	4,963.66
Thursday	195	7,268.30	7,550.00	5,045.59
Friday	199	7,187.53	7,157.00	4,741.61
Saturday	199	7,752.27	6,637.00	6,053.92
Sunday	193	6,606.73	5,600.00	5,641.85

Table A.3 Description: This table summarizes daily step activity across the seven days of the week, including observation counts, average steps, median steps, and standard deviations. Saturday recorded the highest average daily steps (7,752.27), while Sunday recorded the lowest (6,606.73), representing a difference of approximately 1,146 steps. Saturday also exhibited the highest standard deviation (6,053.92), indicating considerable variation in recorded activity. These findings provide additional numerical support for Figure 3 and highlight differences in daily physical activity patterns throughout the week.

Table A.4 — Average Hourly Steps Throughout the Day

HourOfDay	Observations	AverageSteps	MedianSteps	StdSteps	ZeroStepHours	ZeroStepPct
0	1,935.00	43.36	0	147.22	1,331	68.79
1	1,934.00	21.88	0	89.64	1,476	76.32
2	1,934.00	13.69	0	68.1	1,560	80.66
3	1,933.00	6.85	0	39.89	1,647	85.2
4	1,931.00	11.11	0	58.12	1,649	85.4
5	1,931.00	34.93	0	182.32	1,570	81.31
6	1,930.00	148.24	0	413.2	1,199	62.12
7	1,930.00	282.65	3	582.82	959	49.69
8	1,930.00	395.84	54	851.37	806	41.76
9	1,930.00	431.37	117	818.06	665	34.46
10	1,926.00	453.86	171.5	756.99	566	29.39
11	1,924.00	454.74	219	746.37	494	25.68
12	1,918.00	534.26	248.5	880.61	435	22.68
13	1,917.00	496.23	231	789.53	449	23.42
14	1,917.00	506.02	190	922.64	473	24.67
15	1,910.00	398.06	193.5	673.79	464	24.29
16	1,902.00	470.96	229	801.84	443	23.29
17	1,900.00	499.71	258.5	822.87	437	23
18	1,899.00	550.27	282	879.99	440	23.17
19	1,899.00	554.89	223	974.15	444	23.38
20	1,897.00	377.63	156	687.27	491	25.88
21	1,896.00	283.5	68	594.32	641	33.81
22	1,894.00	204.01	26	513.29	778	41.08
23	1,891.00	112.09	0	371.56	1,061	56.11

Table A.4 Description: This table summarizes recorded hourly step activity throughout the 24-hour day, including observation counts, average steps, median steps, standard deviations, and the percentage of observations with zero recorded steps. Average hourly steps were lowest at 3:00 AM (6.85 steps) and generally increased during the morning and daytime hours. Activity reached its highest average at 7:00 PM (554.89 steps), before declining later in the evening. The percentage of observations with zero recorded steps was particularly high overnight, reaching 85.20% at 3:00 AM. These findings provide additional numerical support for Figure 4 and highlight variations in recorded physical activity throughout the day.

Table A.5 — Average Recorded Sleep Duration by Day of Week

DayOfWeek	SleepLogs	Participants	AverageHours	MedianHours	StdHours
Monday	114	18	7.09	7.44	1.91
Tuesday	128	20	6.76	7.15	1.91
Wednesday	134	19	6.83	7.48	2.1
Thursday	122	19	6.93	7.32	1.81
Friday	112	20	6.91	7.2	2.03
Saturday	139	20	6.68	6.97	2.44
Sunday	126	21	7.36	7.92	2.49

Table A.5 Description: This table summarizes recorded sleep duration across the seven days of the week, including sleep-log counts, participant counts, average hours, median hours, and standard deviations. Sunday recorded the longest average sleep duration (7.36 hours), while Saturday recorded the shortest (6.68 hours), representing a difference of approximately 41 minutes. Sunday also exhibited the highest standard deviation (2.49 hours), indicating considerable variation in recorded sleep duration. These findings provide additional numerical support for Figure 5 and highlight differences in recorded sleep patterns throughout the week. Because participant counts and recording coverage varied, these patterns should be interpreted cautiously.

Table A.6 — Correlations Between Daily Activity Measurements

Variable 1	Variable 2	Observations	Pearson	Spearman
TotalSteps	Calories	1,367	0.576	0.550
TotalSteps	VeryActiveMinutes	1,367	0.677	0.754
TotalSteps	FairlyActiveMinutes	1,367	0.354	0.701
TotalSteps	LightlyActiveMinutes	1,367	0.590	0.632
TotalSteps	SedentaryMinutes	1,367	-0.351	-0.457
VeryActiveMinutes	Calories	1,367	0.599	0.519
SedentaryMinutes	Calories	1,367	-0.111	-0.169

Table A.6 Description: This table summarizes Pearson and Spearman correlation coefficients between daily physical activity measurements, based on 1,367 observations. Daily steps demonstrated a positive correlation with very active minutes (Pearson r = 0.677) and estimated calories burned (r = 0.576), indicating that higher step counts were generally associated with greater physical activity intensity and calorie expenditure. Daily steps also showed a negative correlation with sedentary minutes (r = -0.351), while the relationship between sedentary minutes and estimated calories burned was weak (r = -0.111). These findings provide additional numerical support for Figure 6 and demonstrate the value of examining multiple activity measurements when evaluating physical activity patterns. However, these correlations represent statistical associations and do not establish causation.

Appendix B — Additional Analysis and Calculations

This section presents supplementary statistical analyses and calculations performed in Python that provide further context for the main findings. These include additional comparisons, sensitivity analyses, and participant-level relationships that help evaluate the reliability and interpretation of the results.

Table B.1 — Sleep Duration Sensitivity Analysis

Sleep Data	Sleep Logs	Average Duration (Hours)
All recorded sleep logs	1,007	6.34
Excluding flagged unusual durations	875	6.93

Table B.1 Description: This table compares average recorded sleep duration before and after excluding observations flagged for unusual durations. The original dataset contained 1,007 sleep logs with an average recorded duration of 6.34 hours. After excluding 132 flagged logs, the remaining 875 observations had an average duration of 6.93 hours, representing an increase of approximately 0.59 hours (35 minutes). These findings demonstrate that unusual measurements influenced the overall sleep duration estimate and highlight the importance of conducting sensitivity analyses when evaluating data quality. However, excluding flagged observations does not necessarily guarantee that the remaining measurements are accurate.

Table B.2 — Weekday vs. Weekend Activity Comparison

Comparison Method	Avg. Weekday Steps	Avg. Weekend Steps	Difference (Weekend − Weekday)
Observation-Level	7,452.94	7,188.27	-264.67
Participant-Level	7,142.39	6,935.94	-206.45

Table B.2 Description: This table compares average daily step counts between weekdays and weekends using observation-level and participant-level analyses. The observation-level comparison showed an average of 7,452.94 steps on weekdays and 7,188.27 steps on weekends, representing a decrease of 264.67 steps. The participant-level comparison showed an average of 7,142.39 weekday steps and 6,935.94 weekend steps, representing a decrease of 206.45 steps. The participant-level approach gives each participant equal weight, reducing the influence of differences in recorded observation counts. Although both methods indicated slightly lower average activity on weekends, individual patterns varied, suggesting that weekday and weekend activity habits were not consistent across all participants.

Table B.3 — Comparison of Daily Steps and Calorie Correlations

Analysis Method	Pearson Correlation (r)	Interpretation
All Participant-Days	0.576	Moderate positive
Between Participants	0.432	Moderate positive
Within Participants	0.802	Strong positive

Table B.3 Description: This table compares Pearson correlations between daily steps and estimated calorie expenditure using three analytical approaches. The overall participant-day analysis demonstrated a moderate positive correlation (r = 0.576), while the between-participant comparison showed a moderate positive correlation (r = 0.432). The within-participant analysis produced the strongest positive correlation (r = 0.802), indicating that days with higher-than-usual step counts generally corresponded with higher-than-usual estimated calorie expenditure for the same participant. These findings highlight the importance of examining individual activity patterns and demonstrate how different analytical approaches can influence the interpretation of relationships between wellness measurements. However, these correlations do not establish causation.

Appendix C — Python Code and Data Processing Documentation

This section provides selected Python code and documentation supporting the data cleaning, transformation, analysis, and validation procedures performed during the Bellabeat case study. Python was used in Jupyter Notebook, primarily with Pandas and NumPy, to prepare the Fitbit datasets, calculate descriptive statistics, examine activity and sleep patterns, conduct correlation and sensitivity analyses, and export analytical results. The included code excerpts demonstrate key steps in the analytical workflow and support the reproducibility of the findings presented in the report.

C.1 — Data Cleaning and Preparation

Code C.1.1 — Resolving Overlapping Daily Activity Records

```
# Create copies and label their source periods
daily_p1 = daily_check_1.copy()
daily_p2 = daily_check_2.copy()

daily_p1["SourcePeriod"] = "Period 1"
daily_p2["SourcePeriod"] = "Period 2"

# Combine both datasets
daily_combined = pd.concat(
    [daily_p1, daily_p2],
    ignore_index=True
)

# Record the number of observations before cleaning
```

```
rows_before = len(daily_combined)

# Retain Period 2 for overlapping participant-date records
# Period 2 was appended last, so keep='last' gives it priority
daily_combined = daily_combined.drop_duplicates(
    subset=["Id", "ActivityDate"],
    keep="last"
).reset_index(drop=True)
```

Code C.1.1 Description: This Python code combines daily activity records from two Fitbit collection periods and resolves overlapping participant-date observations. The datasets were labeled according to their source periods before consolidation using Pandas. Duplicate participant-date records were removed while prioritizing Period 2, which contained more complete activity measurements for the overlapping date. This procedure reduced the combined dataset from 1,397 records to 1,373 unique participant-date observations, resolving 24 overlapping records without altering the original source files.

Code C.1.2 — Creating Daily Activity Data Quality Flags

```
# Preserve all consolidated records
daily_final = daily_inspection.copy()

# Flag observations with zero calories
daily_final["FlagZeroCalories"] = (
    daily_final["Calories"] == 0
)

# Flag observations with zero steps
daily_final["FlagZeroSteps"] = (
    daily_final["TotalSteps"] == 0
)

# Flag days with fewer than 1440 recorded minutes
daily_final["FlagIncompleteMinutes"] = (
    daily_final["TotalRecordedMinutes"] < 1440
)

# Flag records with zero steps but positive active minutes
active_columns = [
    "VeryActiveMinutes",
    "FairlyActiveMinutes",
    "LightlyActiveMinutes"
]

daily_final["FlagActivityWithoutSteps"] = (
    (daily_final["TotalSteps"] == 0) &
```

```
    (daily_final[active_columns].sum(axis=1) > 0)
)

# Create a separate dataset for calorie analysis
# Exclude zero-calorie observations only
daily_calorie_analysis = daily_final[
    ~daily_final["FlagZeroCalories"]
].copy()
```

Code C.1.2 Description: This Python code creates data quality flags to identify potentially incomplete or inconsistent daily activity measurements without permanently removing observations from the consolidated dataset. The procedure identified 133 records with zero steps, six records with zero calories, 665 records with fewer than 1,440 recorded activity-category minutes, and four records containing positive active minutes despite zero recorded steps. The six zero-calorie records were excluded from a separate calorie-analysis dataset, reducing the observations from 1,373 to 1,367, while the original consolidated dataset was preserved. These quality checks supported more reliable analysis and allowed potentially unusual measurements to be investigated separately.

Code C.1.3 — Creating Time-Based and Sleep Duration Variables

```
# Extract the calendar date from hourly timestamps
hourly_final["ActivityDate"] = (
    hourly_final["ActivityHour"].dt.normalize()
)

# Extract the hour of the day (0–23)
hourly_final["HourOfDay"] = (
    hourly_final["ActivityHour"].dt.hour
)

# Prepare the sleep-log summary
sleep_logs_ready = sleep_logs_final.copy()

# Assign overnight sleep logs to their ending date
sleep_logs_ready["SleepEndDate"] = (
    sleep_logs_ready["EndTime"].dt.date
)

# Convert recorded sleep minutes to hours
sleep_logs_ready["RecordedHours"] = (
    sleep_logs_ready["RecordedMinutes"] / 60
).round(2)
```

Code C.1.3 Description: This Python code creates additional variables to support the analysis of hourly activity and recorded sleep patterns. The ActivityDate and HourOfDay variables were extracted from hourly timestamps to facilitate comparisons across calendar dates and hours of

the day. For sleep records, SleepEndDate was derived from each sleep log's ending timestamp, and RecordedHours was calculated by converting recorded minutes into hours. These transformations helped standardize time-based measurements and prepare the datasets for daily, hourly, and weekly analyses. The resulting variables supported the examination of activity patterns and recorded sleep duration presented in the main report.

C.2 — Statistical Analysis and Calculations

Code C.2.1 — Classifying Participants by Average Daily Steps

```
# Calculate daily step statistics for each participant
participant_activity = (
    daily_activity
    .groupby("Id")
    .agg(
        RecordedDays=("TotalSteps", "count"),
        AverageDailySteps=("TotalSteps", "mean"),
        MedianDailySteps=("TotalSteps", "median")
    )
    .reset_index()
)

# Define the activity group boundaries
step_bins = [
    -np.inf,
    5000,
    7500,
    10000,
    np.inf
]

# Define descriptive activity group labels
step_labels = [
  "Under 5,000",
    "5,000-7,499",
    "7,500-9,999",
    "10,000+"
]

# Assign each participant to an activity group
participant_activity["ActivityGroup"] = pd.cut(
    participant_activity["AverageDailySteps"],
    bins=step_bins,
  labels=step_labels,
    right=False
)
```

```
# Summarize the participant activity groups
activity_group_summary = (
    participant_activity
    .groupby("ActivityGroup", observed=False)
    .agg(
        Participants=("Id", "count"),
        AverageSteps=("AverageDailySteps", "mean")
    )
)

# Calculate each group's percentage of participants
activity_group_summary["Percentage"] = (
    activity_group_summary["Participants"]
    / len(participant_activity)
    * 100
)

# Round results to two decimal places
activity_group_summary = activity_group_summary.round(2)
```

Code C.2.1 Description: This Python code classifies Fitbit participants into four activity groups based on their average daily step counts. Using Pandas, daily activity records were aggregated at the participant level to calculate average and median daily steps. The pd.cut() function was then used to assign participants to predefined step categories: Under 5,000, 5,000–7,499, 7,500–9,999, and 10,000 or more steps. Group-level statistics and percentages were calculated to compare activity patterns across participants. The analysis classified 35 participants and provided the foundation for Figure 1 and the personalized activity recommendations presented in the report.

Code C.2.2 — Calculating Pearson and Spearman Correlations

```
# Select relationships to investigate
variable_pairs = [
    ("TotalSteps", "Calories"),
    ("TotalSteps", "VeryActiveMinutes"),
    ("TotalSteps", "FairlyActiveMinutes"),
    ("TotalSteps", "LightlyActiveMinutes"),
    ("TotalSteps", "SedentaryMinutes"),
    ("VeryActiveMinutes", "Calories"),
    ("SedentaryMinutes", "Calories")
]

# Store correlation results
correlation_results = []
```

```
# Calculate Pearson and Spearman correlations
for variable_1, variable_2 in variable_pairs:

    # Select observations with both measurements
    paired_data = daily_calories[
        [variable_1, variable_2]
    ].dropna()

    # Calculate correlation coefficients
    pearson_corr = paired_data[
        variable_1
    ].corr(
        paired_data[variable_2],
        method="pearson"
    )

    spearman_corr = paired_data[
        variable_1
    ].corr(
        paired_data[variable_2],
        method="spearman"
    )

    # Save the results
    correlation_results.append({
        "Variable 1": variable_1,
        "Variable 2": variable_2,
        "Observations": len(paired_data),
        "Pearson": pearson_corr,
        "Spearman": spearman_corr
    })

# Organize results into a DataFrame
correlation_summary = pd.DataFrame(
    correlation_results
)

# Round correlation coefficients
correlation_summary = correlation_summary.round(3)

# Display results
display(correlation_summary)
```

Code C.2.2 Description: This Python code calculates Pearson and Spearman correlation coefficients to examine relationships between daily physical activity measurements and estimated calorie expenditure. Seven variable pairs were selected for comparison, including

daily steps, activity intensity, sedentary minutes, and calories burned. Missing paired observations were excluded before calculating correlations using the Pandas corr() function. The results were organized into a summary DataFrame and rounded to three decimal places. This analysis provided the statistical foundation for Figure 6 and Table A.6, helping identify positive and negative relationships between activity measurements. The correlations represent statistical associations and do not establish causation.

Code C.2.3 — Sleep Duration Sensitivity Analysis

```
# Identify short, long, and unusually lengthy logs
sleep_flags = [
    "FlagShortLog",
    "FlagLongLog",
    "FlagDuration961"
]

# Identify logs with at least one duration flag
flagged_logs = sleep_logs[sleep_flags].any(axis=1)

print("Total flagged logs:", flagged_logs.sum())
print("Unflagged logs:", (~flagged_logs).sum())

# Compare original and unflagged sleep durations
sleep_comparison = pd.DataFrame({
    "All Sleep Logs": sleep_logs["RecordedHours"].agg(
        ["count", "mean", "median", "std"]
    ),
    "Unflagged Sleep Logs": sleep_logs.loc[
        ~flagged_logs, "RecordedHours"
    ].agg(["count", "mean", "median", "std"])
})

# Round results to two decimal places
sleep_comparison = sleep_comparison.round(2)

display(sleep_comparison)
```

Code C.2.3 Description: This Python code performs a sensitivity analysis to evaluate how unusually short and long sleep logs influence recorded sleep duration statistics. Three existing data quality flags were used to identify observations with potentially unusual durations. Of the 1,007 sleep logs, 132 were flagged, leaving 875 unflagged observations. Excluding flagged records increased the average recorded sleep duration from 6.34 to 6.93 hours and reduced the standard deviation from 2.83 to 2.13 hours. These findings demonstrate how unusual observations can influence descriptive statistics and highlight the importance of evaluating data quality before interpreting analytical results. Flagged observations were not automatically

considered invalid, and the results represent recorded sleep-log durations rather than confirmed time asleep.

C.3 — Data Validation and Export

Code C.3.1 — Exporting Analytical Summary Tables to CSV

```
# Define the folder for analytical outputs
output_dir = project_dir / "Analysis_Outputs"

# Create the folder if it does not already exist
output_dir.mkdir(parents=True, exist_ok=True)

# Organize the summary tables for export
summary_exports = {
    "weekday_steps.csv": weekday_summary.reset_index(),
    "hourly_steps.csv": hourly_step_summary.reset_index(),
    "hourly_intensity.csv": hourly_intensity_summary.reset_index(),
    "hourly_calories.csv": hourly_calorie_summary.reset_index(),
    "sleep_by_weekday.csv": sleep_weekday_summary.reset_index(),
    "activity_groups.csv": activity_group_summary.reset_index(),
    "activity_group_profiles.csv":
activity_profile_summary.reset_index(),
    "daily_activity_correlations.csv": correlation_summary,
    "key_findings.csv": key_findings,
    "business_insights.csv": business_insights
}

# Export each summary table as a CSV file
for filename, df in summary_exports.items():

    file_path = output_dir / filename

    # Save without the Pandas row index
    df.to_csv(file_path, index=False)

    print(f"Exported: {filename} ({len(df)} rows)")

# Confirm the number of exported files
print("\nTotal files exported:", len(summary_exports))
```

Code C.3.1 Description: This Python code exports analytical summary tables generated during the Bellabeat Analyze Phase into CSV files for visualization, reporting, and documentation. Using Pandas and pathlib, an Analysis_Outputs directory was created to organize the exported results. Ten summary DataFrames containing daily and hourly activity statistics, sleep patterns, participant activity groups, correlation analyses, and business insights were organized into a dictionary and exported using the to_csv() function. The DataFrame indexes were excluded

from the exported files, while relevant grouping variables were preserved as columns. This procedure produced 10 CSV files that supported the Tableau visualizations, statistical tables, and findings presented in the final report.

Code C.3.2 — Validating Exported CSV Files

```
# Store verification results
validation_results = []

# Verify each exported CSV
for filename, original_df in summary_exports.items():

    # Reload the exported file
    file_path = output_dir / filename
    reloaded_df = pd.read_csv(file_path)

    # Check row counts and column names
    rows_match = len(reloaded_df) == len(original_df)

    columns_match = (
        reloaded_df.columns.tolist()
        == original_df.columns.tolist()
    )

    # Compare exported values with original values
    try:
        pd.testing.assert_frame_equal(
            original_df.reset_index(drop=True),
            reloaded_df,
            check_dtype=False,
            check_categorical=False,
            check_exact=False,
            rtol=1e-8,
            atol=1e-8
        )
        values_match = True

    except AssertionError:
        values_match = False

    # Assign validation status
    status = (
        "PASS"
        if rows_match and columns_match and values_match
        else "REVIEW"
```

```
    )

    # Record verification results
    validation_results.append({
        "File": filename,
        "Expected Rows": len(original_df),
        "Actual Rows": len(reloaded_df),
        "Columns Match": columns_match,
        "Values Match": values_match,
        "Status": status
    })

# Create and display validation summary
validation_summary = pd.DataFrame(validation_results)

display(validation_summary)

# Count successfully verified files
passed_files = (
    validation_summary["Status"] == "PASS"
).sum()

print(
    f"Files successfully verified: "
    f"{passed_files} of {len(summary_exports)}"
)
```

Code C.3.2 Description: This Python code validates exported analytical summary tables by reloading each CSV file into Pandas and comparing it with its original DataFrame. The validation procedure checks row counts, column names, and numerical values using the pd.testing.assert_frame_equal() function with specified comparison tolerances. Each file receives a validation status based on whether its contents match the expected results. All 10 exported CSV files passed verification, with no discrepancies identified within the specified tolerances. This procedure helped ensure the integrity and consistency of the analytical outputs used for Tableau visualizations, supporting tables, and the final report.